

Ordinals

Well-founded relations, well-orders, von Neumann ordinals, Burali-Forti, and the representation theorem

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We work in ZFC, but — as noted below — we deliberately avoid using **Foundation**, so that “being an ordinal” does real work.

Why count past $\mathbb{N}$?

Cantor was led to the transfinite by a concrete problem. Given $A \subseteq \mathbb{R}$, its **derivative** is the set of its limit points,

$$A' = \{x \in A : x \text{ is a limit point of } A\}.$$

Iterating gives a descending chain $A \supseteq A' \supseteq A'' \supseteq \dots$, and one can take a “limit” $A^{(\infty)} = \bigcap_n A^{(n)}$. But $A^{(\infty)}$ may still have isolated points, so the process continues: $A^{(\infty)} \supsetneq (A^{(\infty)})' \supsetneq \dots$. To index the stages we need numbers *beyond* the natural numbers:

$$0, 1, 2, \dots, \omega, \omega + 1, \omega + 2, \dots, \omega + \omega, \dots, \omega \cdot \omega, \dots$$

Making “keep counting, even through limit stages” precise is exactly what the **ordinals** accomplish. The concept they rest on is the **well-order**.

Well-founded relations

Throughout, R is a relation on a set A , and for $X \subseteq A$ we write $R \upharpoonright X = \{\langle x, y \rangle \in R : x, y \in X\}$.

y is R -minimal in X : $y \in X$ and $\neg \exists z (z \in X \wedge z R y)$ — nothing in X is R -below y .

R is well-founded on A : every nonempty $X \subseteq A$ has an R -minimal element:

$$\forall X (\emptyset \neq X \subseteq A \rightarrow \exists y \in X \forall z \in X \neg(z R y)).$$

Two basic facts:

- The **Axiom of Foundation** says exactly that $\in$ is well-founded on every set. We will *not* assume this below.
- If A is **finite**, then R is well-founded on A iff R has **no cycles** $a_0 R a_1 R \cdots R a_n R a_0$ on A . (For infinite A this fails: the usual $<$ on $\mathbb{Q}$ is acyclic but not well-founded.)

Well-foundedness is the “no infinite descent” condition; it is what makes proof by induction and definition by recursion possible.

Well-orders

Definition. R is a **well-order (WO)** on A : R totally orders A strictly *and* R is well-founded on A .

Equivalently: a strict total order in which **every nonempty subset has a least element**. (In a total order an R -minimal element of X is automatically the R -least element of X .)

Examples.

- $(\mathbb{N}, <)$ is a well-order — this is the least number principle.
- $(\{1 - 2^{-n} : n \in \mathbb{N}\}, <)$ is a well-order.
- $(\{m - 2^{-n} : m, n \geq 1\}, <)$ is a well-order (a copy of $\mathbb{N}$ followed by another copy, then another, ...).
- $(\mathbb{Q}, <)$, $(\mathbb{R}, <)$, $(\mathbb{Z}, <)$ are **not** WOs.

Well-orders are rigid

Orders are compared by **embeddings**: $f : (A, <_A) \rightarrow (B, <_B)$ with $x <_A y \leftrightarrow f(x) <_B f(y)$. An **isomorphism** is a bijective embedding; $\mathbb{Z}$ and $\mathbb{R}$ have nontrivial automorphisms ($z \mapsto z + 1$). Well-orders do not.

Proposition. Let $(A, <)$ be a well-order and $f : A \rightarrow A$ a self-embedding. Then $x \leq f(x)$ for all $x \in A$.

Proof. If $\{x \in A : f(x) < x\} \neq \emptyset$, it has a least element z . Then $f(z) < z$, and applying the embedding $f(f(z)) < f(z)$ — so $f(z)$ is a smaller element of the set, contradicting minimality of z . $\square$

Corollary. The only automorphism of a well-order is the identity.

Proof. If f is an automorphism, both f and f^{-1} are self-embeddings, so $x \leq f(x)$ and $x \leq f^{-1}(x)$; the latter gives $f(x) \leq x$. $\square$

For $b \in A$, the **initial segment below** b is

$$A_{<b} := \{a \in A : a < b\}.$$

Corollary. No well-order is isomorphic to a proper initial segment of itself.

Proof. Suppose $f : A \xrightarrow{\cong} A_{<b}$. Then $f(x) < b$ for every x , in particular $f(b) < b$ — contradicting the Proposition. $\square$

Ordinal numbers

Cantor defined an ordinal as an *isomorphism class of well-orders* (an “order type”). **von Neumann** gave a canonical representative of each class: a specific set on which $\in$ is the ordering. For instance

$$3 = \{ \emptyset, \{ \emptyset \}, \{ \emptyset, \{ \emptyset \} \} \}$$

is a 3-element set well-ordered by $\in$.

Definition. A set A is transitive: $\forall x \in A (x \subseteq A)$ — every element is also a subset. (A transitive set cannot “hide” members inside its members.)

Definition (von Neumann). A set z is an ordinal, $\text{Ord}(z)$: z is transitive and well-ordered by $\in$.

We do **not** assume Foundation here. If we did, “well-ordered by $\in$ ” could be weakened to “*linearly* ordered by $\in$ ”, since well-foundedness would come for free.

Proposition 1. If $\text{Ord}(\alpha)$ and $x \in \alpha$, then $\text{Ord}(x)$. (*Every element of an ordinal is an ordinal.*)

Proof. Since α is transitive, $x \subseteq \alpha$, so $\in$ well-orders x (a subset of a well-order is well-ordered). It remains to show x is a transitive set. Let $y \in z \in x$. Then $z \in \alpha$ (as $x \subseteq \alpha$), so $y \in \alpha$ (as $z \subseteq \alpha$). Now $x, y \in \alpha$ and $\in$ is a total order on α , so

$$y \in x \quad \vee \quad x \in y \quad \vee \quad x = y.$$

If $x = y$ then $x \in z \in x$; if $x \in y$ then $x \in y \in z \in x$ — both are $\in$ -cycles in α , impossible since $\in$ is well-founded on α . Hence $y \in x$. $\square$

This suggests ordering ordinals by

$$\alpha < \beta \quad :\Longleftrightarrow \quad \alpha \in \beta,$$

and Proposition 1 gives, for every ordinal,

$$\alpha = \{ \beta : \beta < \alpha \}$$

— an ordinal is the set of all smaller ordinals.

The well-ordering of the ordinals

Proposition 2. For ordinals α, β : $\alpha < \beta \iff \alpha \subsetneq \beta$.

Proof. $(\Rightarrow)$ $\alpha \in \beta$ and β transitive give $\alpha \subseteq \beta$; properly, since $\alpha \notin \alpha$ (well-foundedness) while $\alpha \in \beta$.

$(\Leftarrow)$ We prove the stronger statement

$$\text{trans}(a) \wedge a \subsetneq \beta \implies \text{Ord}(a) \wedge a \in \beta.$$

If $a \subsetneq \beta$ then a is linearly ordered by $\in$ (subset of β); being also transitive, a is an ordinal. For $a \in \beta$: since $\beta \setminus a \neq \emptyset$, pick $\gamma \in$ -minimal in $\beta \setminus a$. We claim $\gamma = a$.

- $\gamma \subseteq a$: if $g \in \gamma$ then $g \notin \beta \setminus a$ by minimality, and $g \in \beta$ (transitivity of β), so $g \in a$.
- $a \subseteq \gamma$: let $x \in a$. Then $x \in \beta$, so by trichotomy in β , $x \in \gamma \vee \gamma \in x \vee x = \gamma$. The last two force $\gamma \in a$ (transitivity of a , resp. directly), contradicting $\gamma \notin a$. So $x \in \gamma$.

Hence $a = \gamma \in \beta$. $\square$

Write $ON = \{ x : \text{Ord}(x) \}$ for the collection of all ordinals.

Theorem. ON is well-ordered by $< (= \in)$.

Since ON is not a set (next result), this is shorthand for a list of ZFC theorems: $\in$ on the ordinals is (1) transitive, (2) irreflexive, (3) trichotomous, and (4) well-founded — *every nonempty set of ordinals has an $\in$ -least member*.

Proof.

1. **Transitive:** $\alpha < \beta < \gamma$ means $\alpha \in \beta \in \gamma$; γ transitive gives $\alpha \in \gamma$.
2. **Irreflexive:** $\alpha \in \alpha$ would be an $\in$ -cycle in α , contradicting well-foundedness of $\in$ on α .
3. **Trichotomy:** first a lemma.

Lemma. $\alpha \cap \beta$ is an ordinal.

It is transitive (an intersection of transitive sets is transitive) and well-ordered by $\in$ (a subset of α).

Let $\gamma = \alpha \cap \beta$, so $\gamma \subseteq \alpha$ and $\gamma \subseteq \beta$. By Proposition 2, $\gamma \in \alpha$ or $\gamma = \alpha$, and likewise for β . If $\gamma \in \alpha$ and $\gamma \in \beta$ then $\gamma \in \alpha \cap \beta = \gamma$, impossible. So $\gamma = \alpha$ or $\gamma = \beta$; that is $\alpha \subseteq \beta$ or $\beta \subseteq \alpha$, hence (Proposition 2) $\alpha \leq \beta$ or $\beta \leq \alpha$.

4. **Well-founded:** let $A \neq \emptyset$ be a set of ordinals and $\alpha \in A$. If α is $\in$ -least, done. Otherwise $\alpha \cap A = \{\beta \in A : \beta < \alpha\}$ is a nonempty subset of α , so it has an $\in$ -least element (since $\in$ well-orders α) — and that is $\in$ -least in A . $\square$

Burali-Forti. ON is a **proper class**: no set contains all the ordinals.

Proof. If every ordinal were in a set X , then $ON = \{y \in X : \text{Ord}(y)\}$ would be a set. It is transitive (Proposition 1) and well-ordered by $\in$ (Theorem), hence an ordinal, hence $ON \in ON$ — contradicting irreflexivity. (This needs no Foundation: ordinals are *by definition* strictly ordered by $\in$.) $\square$

Working in Cantor's naive set theory, Burali-Forti (1897) put together “the well-ordering of all well-orderings” to get a well-ordering strictly longer than every well-ordering, including itself.

The natural numbers, again

Recall ω was introduced (Axiom of Infinity) as the **smallest inductive set** — the smallest set containing $\emptyset$ and closed under the successor

$$S(x) = x \cup \{x\}.$$

Proposition 3. If $\text{Ord}(\alpha)$ then $\text{Ord}(S(\alpha))$, and for every ordinal γ , $\gamma < S(\alpha) \iff \gamma \leq \alpha$.

Proof (sketch). $S(\alpha) = \alpha \cup \{\alpha\}$ is transitive: its members are the members of α (each $\subseteq \alpha \subseteq S(\alpha)$) and α itself ($\subseteq S(\alpha)$). It is well-ordered by $\in$: it is α with one new top element. The equivalence is then immediate from $\gamma \in \alpha \cup \{\alpha\}$. $\square$

α **is a successor** $\alpha = S(\beta)$ for some ordinal β .

α **is a limit** $\alpha \neq 0$ and α is not a successor. Equivalently: $\alpha \neq 0$ and $\forall \beta < \alpha (S(\beta) < \alpha)$.

α **is finite (a natural number)** every $\beta \leq \alpha$ is either 0 or a successor.

Proposition 4. If α is finite, so is $S(\alpha)$.

Now consider the class $X = \{x : x \text{ is a finite ordinal}\}$. It contains 0 and is closed under S (Proposition 4) — it is **inductive**. Since ω is the *smallest* inductive set, $\omega \subseteq X$: every element of ω is a finite ordinal.

Claim: there is no finite ordinal outside ω . Suppose $n \in X \setminus \omega$. (We cannot apply the least-element principle to $X \setminus \omega$ directly — it might be a proper class.) But $S(n) \setminus \omega$ is a *set* (a subset of the ordinal $S(n)$), nonempty (n is in it), and consists of finite ordinals. So it has an $\in$ -least element k .

- $k \neq 0$, since $0 \in \omega$.
- So k is a successor, $k = S(j)$. Then $j \in k$, so by leastness $j \notin S(n) \setminus \omega$; but $j \in S(n)$, so $j \in \omega$. Since ω is closed under S , $k = S(j) \in \omega$ — contradicting $k \notin \omega$.

So $X \setminus \omega = \emptyset$. $\square$

Proposition 5. Every set X that is an **initial segment of ON** — i.e. $\forall \beta \in X \forall \alpha < \beta (\alpha \in X)$ — is an ordinal.

Proof. X is well-ordered by $\in$ (Theorem). The initial-segment condition says $\alpha \in \beta \in X \Rightarrow \alpha \in X$, i.e. X is transitive. $\square$

Corollary. ω is an ordinal.

Proof. By the Claim, ω is a set of ordinals; and it is an initial segment of ON (if $\beta \in \omega$ and $\alpha < \beta$ then α is a finite ordinal, and $\alpha \in \omega$ because every element of a finite ordinal is a finite ordinal, hence in ω). Apply Proposition 5. $\square$

Corollary. ω is the least limit ordinal.

Proof. $\omega \neq 0$, and ω is not a successor (if $\omega = S(n)$ then $n \in \omega$ and $\omega = S(n) \in \omega$). So ω is a limit. Every smaller ordinal is a finite ordinal, hence 0 or a successor — not a limit. $\square$

We can keep going:

$$\omega, \quad S(\omega) = \omega + 1, \quad \omega + 2, \quad \dots, \quad \underbrace{\omega + \omega}_{\omega \cdot 2}, \quad \dots, \quad \omega \cdot \omega, \quad \dots, \quad \omega^\omega, \quad \dots, \quad \omega^{\omega^\omega}, \quad \dots, \quad \varepsilon_0, \quad \dots$$

(Ordinal $+$, $\cdot$, and exponentiation are defined by transfinite recursion later.) Every ordinal displayed here is still a *countable* set; the supremum of all countable ordinals is ω_1 , the first uncountable ordinal.

Representing well-orders as ordinals

The point of the von Neumann definition: each well-order should have exactly one ordinal representative.

We have already shown, for ordinals:

1. no ordinal is isomorphic to a proper initial segment of itself;
2. the only automorphism of an ordinal is the identity.

By trichotomy, (1) gives **uniqueness**: if $\alpha \cong \beta$ then $\alpha = \beta$. (If $\alpha < \beta$ then $\alpha = \beta_{<\alpha}$ is a proper initial segment of β , so $\alpha \cong \beta$ would violate (1).)

Representation Theorem for WOs. If R well-orders A , then there is a **unique** ordinal α with $(A, R) \cong (\alpha, \in)$. The isomorphism is unique.

α is called the **order type** of (A, R) , written $\text{type}(A, R)$.

Proof. Uniqueness of α is the remark above; uniqueness of the isomorphism follows from (2). For existence:

For $a \in A$ put $A_{<a} = \{x \in A : x R a\}$; it is well-ordered by R . Call a **good** if $(A_{<a}, R) \cong (\xi, \in)$ for some ordinal ξ — by uniqueness this ξ and the isomorphism h_a are then determined. Let

$$G = \{a \in A : a \text{ is good}\}.$$

By **Replacement**, $f : G \rightarrow ON$, $f(a) =$ the unique ordinal $\cong (A_{<a}, R)$, is a function.

Observation. For $a \in G$ and $c R a$: $c \in G$, $h_c = h_a \upharpoonright A_{<c}$, and $f(c) = h_a(c)$. In particular G is an initial segment of A , and f is order-preserving: $a R b \Rightarrow f(a) = h_b(a) \in f(b)$, i.e. $f(a) < f(b)$.

$\text{ran}(f)$ is an initial segment of ON . If $\xi = f(a) = \text{ran}(h_a)$ and $\eta < \xi$, then $\eta = h_a(b)$ for some $b R a$, and $h_a(b) = f(b)$, so $\eta \in \text{ran}(f)$.

By Proposition 5, $\alpha := \text{ran}(f)$ is an ordinal, and $f : G \xrightarrow{\cong} \alpha$.

Finally $G = A$. If not, let a be R -minimal in $A \setminus G$. Since G is an initial segment of A , $A_{<a} = G$, so $(A_{<a}, R) \cong (\alpha, \in)$ — making a good, a contradiction. $\square$

Looking ahead: transfinite induction

The existence proof quietly used a “least counterexample” argument on a well-order. Its systematic form, valid on ON itself:

Transfinite Induction. Let P be a property of ordinals. If

- $P(0)$;
- $P(\alpha) \Rightarrow P(S(\alpha))$ for all α ;
- for every limit λ : $(\forall \beta < \lambda P(\beta)) \Rightarrow P(\lambda)$;

then $P(\alpha)$ for all ordinals α .

Informally: a least α with $\neg P(\alpha)$ would have to be 0, a successor, or a limit, and each case is excluded. Turning this — and the matching **recursion** principle — into theorems (given that ON is a proper class) is the next topic.